Form B

1. (a) $P(B^c) = 1 - P(B) = 1 - 0.6 = 0.4$

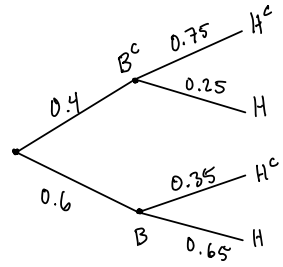
(b) $P(H \mid B) = 0.65$

(c) $P(H) = P(H \text{ and } B) + P(H \text{ and } B^c)$

$= P(H \mid B)\,P(B) + P(H \mid B^c)\,P(B^c)$

$= (0.65)(0.6) + (0.25)(0.4)$

$= 0.49$



(d) $P(B \text{ and } H) = P(H \mid B)\,P(B) = (0.65)(0.6) = 0.39$

Since this probability is nonzero, $B \cap H \neq \emptyset$. B, H are not mutually exclusive.

(e) $P(H \mid B) = 0.65$, different from $P(H) = 0.49$. B, H are not independent.

2. (a), (c) and (f)

3. (a) IV (b) III (c) VI (d) II

4. (a) $X \sim \text{Binom}(10, 0.25)$

(b) $\mu_X = np = (10)(0.25) = 2.5$

(c) $P(X=2) = {}_{10}C_2\,(0.25)^2(0.75)^8 = \dfrac{10!}{8!\,2!}(0.0625)(0.1001)$

$= \dfrac{(10)(9)}{2}(0.0625)(0.1001) = 0.2815$

5. (a) German & Japanese-made cars in the U.S.

(b)

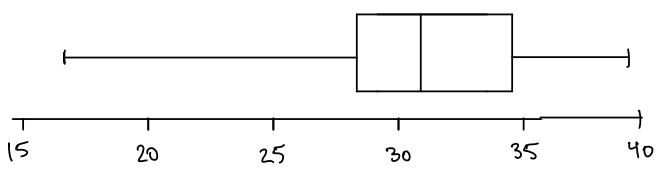
Variables	Type
Country of origin	categorical
Bright green or not?	categorical

6. (a) unimodal, left-skewed

(b) $q_1 = 28.5$, $q_3 = 34.5$ (estimated)

(c) 16.5, 28.5, 31, 34.5, 39.5

(d)



(e) $iqr = 34.5 - 28.5 = 6$

$1.5 \times iqr = (1.5)(6) = 9$

values smaller than $28.5 - (1.5)\,iqr = 19.5$ are flagged

(f) The median is larger,

7. (a) $P(X = 11) = 1 - (0.2 + 0.25 + 0.32) = 0.23$

(b) $P(X \geq 5) = P(X=5) + P(X=7) + P(X=11) = 0.25 + 0.32 + 0.23 = 0.8$

(c) $\mu_X = (3)(0.2) + (5)(0.25) + (7)(0.32) + (11)(0.23) = 6.62$

8. (a) names(dogs)

(b) nrow(dogs)

(c) mean(~ weight, data = dogs)